

Mock validation of the separation-space Gaussian covariance of windowed multi-tracer power-spectrum multipoles

thecov validation note — run results_v5 (mock version 4) · 30 September 2026

Abstract. We compare the Gaussian covariance of power-spectrum multipoles computed by `thecov` — a separation-space (tripolar) formulation that includes the survey window exactly and approximates only Gaussianity, a local plane-parallel line of sight and a window slowly varying over a correlation length — with the sample covariance of 1000 Gaussian mocks of a two-tracer survey with a realistic, wide-angle footprint. For the 174-element data vector ($\ell = 0, 2$ of the two auto- and one cross-spectrum, $0.01 < k < 0.30 h \text{ Mpc}^{-1}$) the mean χ^2 computed with the analytic matrix is 174.26 against an expectation of 173.83 ± 0.59 ($+0.7\sigma$); its variance, its distribution, the eigenvalues of $C^{-1/2} \hat{C} C^{-1/2}$ and the correlation-coefficient residuals are all consistent with an exact covariance, and every block agrees within 1.5σ . A control run that subtracts the *expected* instead of the *realised* shot noise fails at $+7.1\sigma$ in the high- k monopole-auto blocks, a failure reproduced without free parameters by the Poisson fluctuation of the self-pair sum. The only departure from exactness we can see is a variance deficit of $1.9 \pm 1.5\%$ in the three lowest k bins, of the sign expected from the unmodelled integral constraint.

1. Purpose

`thecov` computes the disconnected covariance of FKP–Yamamoto power-spectrum multipoles as

$$C_{\ell_1 \ell_2}^{ABCD}(i, j) = \frac{(4\pi)^4}{I_{AB} I_{CD}} \sum_{L_1 L_2} \frac{1}{(2L_1 + 1)(2L_2 + 1)} \sum_{\lambda \lambda'} (-1)^{(\lambda + \lambda')/2} \sum_{\Lambda_1 \Lambda_2 \Lambda} \left[t^{(1)} \mathcal{I}_{ij}^{(1)} + t^{(2)} \mathcal{I}_{ij}^{(2)} \right], \quad (1)$$

where $\mathcal{I}_{ij}^{(n)} = \int s^2 ds \bar{p}_{L_1 \lambda}^{(i)}(s) \bar{p}_{L_2 \lambda'}^{(j)}(s) \mathcal{Q}_{\Lambda_1 \Lambda_2 \Lambda}^{\omega \omega'}(s)$ couples bin-averaged Bessel transforms of the model multipoles to tripolar window functions $\mathcal{Q}(s)$ measured from pair counts of the random catalogues. Every other test in the suite checks the implementation against exact identities; the mocks are the only test of the three *approximations* behind Equation 1. Because each mock galaxy is displaced along its own line of sight inside a 30° cap, nothing in their construction assumes a plane-parallel geometry.

2. Set-up

Survey. A spherical cap of half-opening 30° between comoving distances 400 and $900 h^{-1} \text{ Mpc}$, with three circular holes and a linear completeness gradient of $\pm 25\%$ across the cap (Figure 1). Two tracers share the angular mask but have different radial selections: $\bar{n}_A$ peaks at $6 \times 10^{-4} h^3 \text{ Mpc}^{-3}$ at $550 h^{-1} \text{ Mpc}$, $\bar{n}_B$ at 3×10^{-4} at $750 h^{-1} \text{ Mpc}$. The expected numbers of galaxies are 63 800 (A) and 47 500 (B); each tracer has its own random catalogue, $15 \times$ denser ($\alpha = 0.0667$). FKP weights $w = (1 + \bar{n} P_0)^{-1}$ with $P_0 = 10^4 h^{-3} \text{ Mpc}^3$.

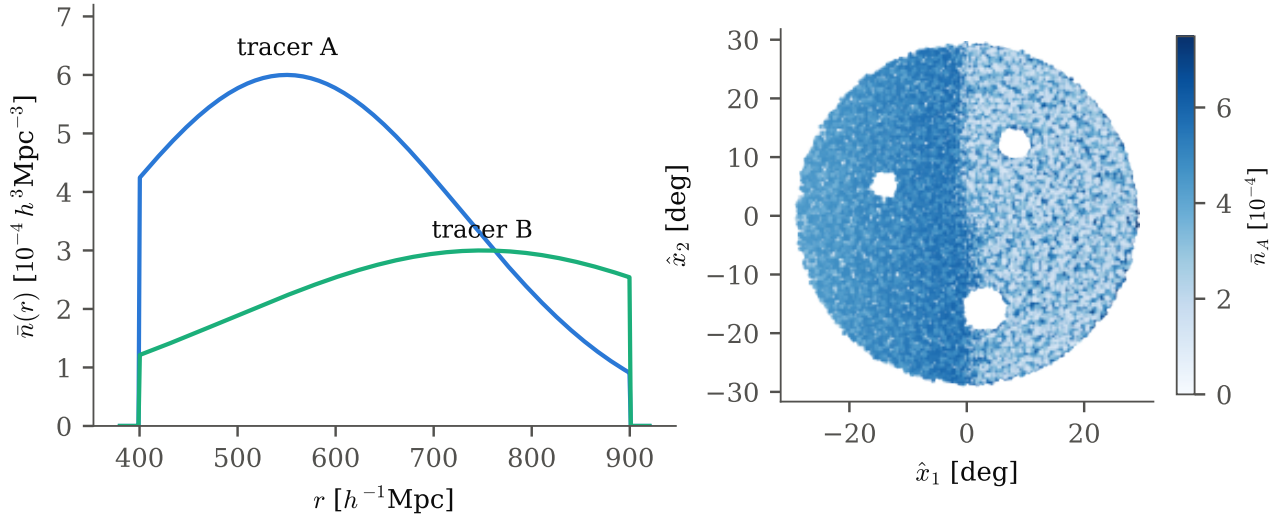


Figure 1: Survey geometry. *Left*: radial selection of the two tracers. *Right*: the angular footprint, shown by a subsample of the randoms of tracer A coloured by $\bar{n}_A$; the three holes and the completeness gradient are visible.

Mocks. A Gaussian linear density field with a smooth, CDM-like spectrum with a small oscillatory feature, generated on a 256^3 grid in a periodic box of side $1800 h^{-1} \text{ Mpc}$ (twice the survey extent; cell $7.03 h^{-1} \text{ Mpc}$,

$k_{\text{Nyq}} = 0.447 h \text{ Mpc}^{-1}$). Galaxies are Poisson-sampled cell by cell from $\bar{n}(1 + b\delta)$ with $b_A = 1.9$, $b_B = 1.2$, and displaced by the linear Zel'dovich velocity $f(\Psi \cdot \hat{x})\hat{x}$, $f = 0.78$, along *each galaxy's own* line of sight. The amplitude is scaled so that $\sigma(b_A\delta) = 0.31$ per cell, for which clipping of $1 + b\delta$ at zero affects 6×10^{-4} of the cells and removes $\approx 0.1\%$ of the power. The field is generated with the cell window pre-divided, so the sampled point process has exactly the input spectrum, and has no power above $0.8 k_{\text{Nyq}}$. The survey mask and weights are applied at the observed (redshift-space) position. The expected multipoles are the linear Kaiser ones, $P_{XY}(k, \mu) = (b_X + f\mu^2)(b_Y + f\mu^2)P_{\text{lin}(k)}$.

Estimator. Yamamoto-FKP multipoles with the line of sight at the first field, TSC assignment with interlacing, and the pypower/jaxpower conventions: α and the shot noise $\sum_g w_g^2 + \alpha^2 \sum_r w_r^2$ are those of each realisation. We measure P_ℓ^{AA} , P_ℓ^{AB} and P_ℓ^{BB} for $\ell = 0, 2$ in 29 bins of width 0.01 between $k = 0.01$ and 0.30 ($k_{\text{max}} = 0.67 k_{\text{Nyq}}$; a dedicated aliasing test on a common catalogue bounds the aliasing bias of this estimator by 0.2% at $0.56 k_{\text{Nyq}}$ and by 0.4% (0.9% for AB) at $0.73 k_{\text{Nyq}}$). The normalisation is fixed to $I_{XY} = \int \bar{n}_X \bar{n}_Y w_X w_Y$. Figure 2 shows the mean of the 1000 mocks.

Analytic covariance. thecov with the Kaiser multipoles above as the model (up to $L = 4$), window functions from pair counts of 5000 (all pairs, $s > 80 h^{-1} \text{ Mpc}$) and 2×10^5 (neighbours, $s < 80$) randoms per window, $\Delta s = 10 h^{-1} \text{ Mpc}$ for the pair counts and $2 h^{-1} \text{ Mpc}$ for the radial integrals. Figure 7 shows representative window functions. The model is exact for the mocks at linear order; the Zel'dovich displacement of a discrete sample adds small non-linear corrections.

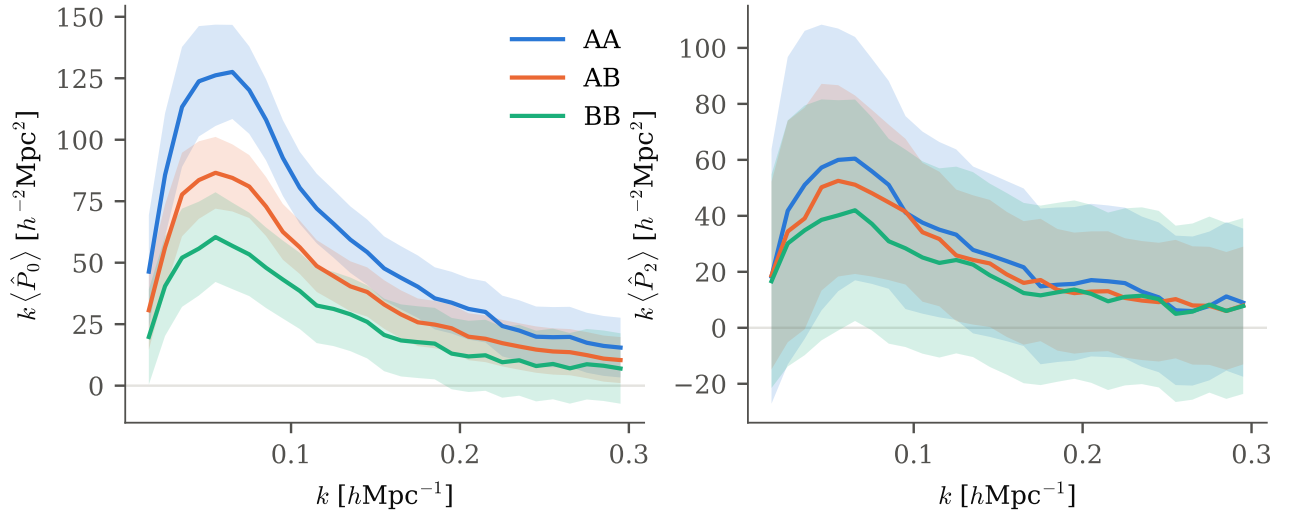


Figure 2: Mean of the 1000 mocks, $kP_{\ell(k)}$, for the monopole (left) and quadrupole (right) of the three spectra. Bands: $\pm 1\sigma$ of a single realisation from the analytic covariance.

3. Statistics

With $N = 1000$ mocks $\mathbf{d}_i$ and $n = 174$ elements, the principal whole-matrix test is

$$\chi_i^2 = (\mathbf{d}_i - \bar{\mathbf{d}})^T C^{-1} (\mathbf{d}_i - \bar{\mathbf{d}}), \quad \langle \chi^2 \rangle = n(1 - 1/N) \pm n(1 - 1/N) \sqrt{2/(nN)}, \quad (2)$$

which, with C exact, is accurate to 0.34% here and sensitive to the whole correlation structure. Its variance should be $\approx 2n$ and its distribution χ_n^2 rescaled by $1 - 1/N$. The eigenvalues of $C^{-1/2} \hat{C} C^{-1/2}$, with $\hat{C}$ the sample covariance, are *not* expected to lie within $\pm \sqrt{2/N}$ of unity: for an exact C they fill the Marchenko–Pastur interval $[(1 - \sqrt{q})^2, (1 + \sqrt{q})^2]$ with $q = n/(N - 1)$, i.e. $[0.339, 2.009]$. To localise a failure we also evaluate Equation 2 restricted to each block and to its low- and high- k halves (split at $k = 0.155$), the ratio $\sigma_{\text{mock}}/\sigma_{\text{thecov}}$ per element (statistical scatter $1/\sqrt{2(N - 1)} = 0.022$) and the correlation-coefficient residuals $z_{ij} = (\hat{\rho}_{ij} - \rho_{ij})\sqrt{N - 1}/(1 - \rho_{ij}^2)$, which are unit Gaussians for an exact C .

4. Results

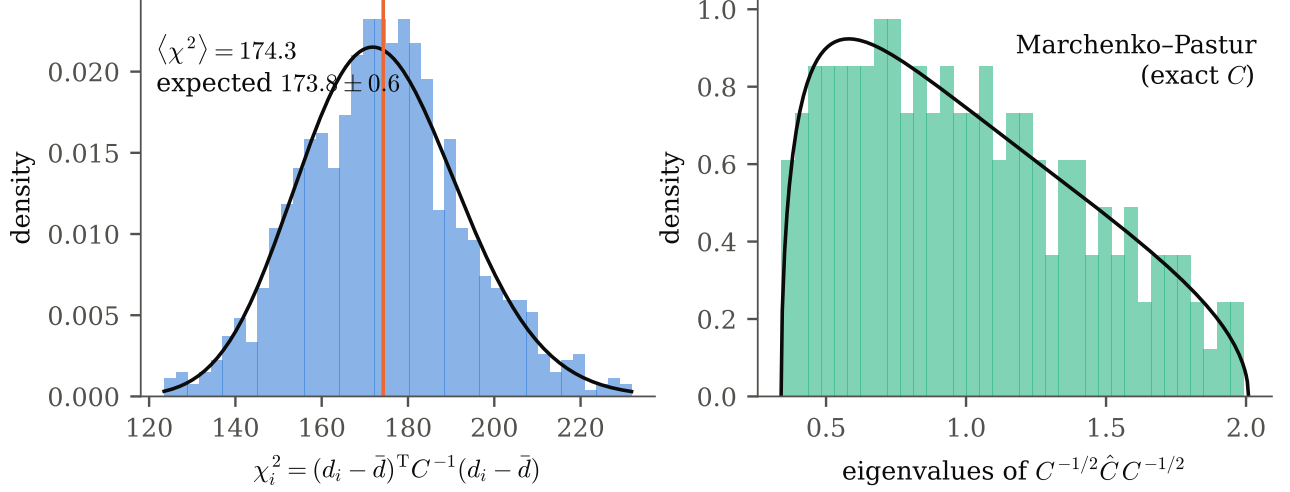


Figure 3: *Left*: distribution of χ_i^2 over the mocks against the rescaled χ_{174}^2 density. *Right*: eigenvalues of $C^{-1/2} \hat{C} C^{-1/2}$ against the Marchenko–Pastur density of an exact covariance with $n = 174$, $N = 1000$.

Whole matrix. $\langle \chi^2 \rangle = 174.26$ against 173.83 ± 0.59 ($+0.7\sigma$); the variance is 345.6 against 347.7, and a Kolmogorov–Smirnov test of the χ_i^2 against the rescaled χ_{174}^2 distribution gives $p = 0.30$ (Figure 3, left). The eigenvalues span $[0.342, 1.991]$, inside the Marchenko–Pastur interval $[0.339, 2.009]$, and their density follows it (Figure 3, right): there is no direction in data space along which the mocks fluctuate more, or less, than thecov predicts, beyond what 1000 samples allow.

Blocks. Table 1 lists Equation 2 restricted to each block. All 18 entries are within 1.53σ ; their spread is what 18 unit Gaussians give.

block	all k	$k < 0.155$	$k > 0.155$
$AA, \ell = 0$	0.994 (−0.7)	0.988 (−1.0)	1.002 (+0.1)
$AA, \ell = 2$	1.008 (+0.9)	0.997 (−0.2)	1.018 (+1.5)
$AB, \ell = 0$	0.995 (−0.6)	1.002 (+0.1)	0.990 (−0.9)
$AB, \ell = 2$	0.995 (−0.6)	0.995 (−0.4)	0.994 (−0.5)
$BB, \ell = 0$	1.001 (+0.1)	0.987 (−1.1)	1.014 (+1.2)
$BB, \ell = 2$	1.000 (0.0)	1.015 (+1.2)	0.987 (−1.1)

Table 1: $\langle \chi^2 \rangle$ restricted to each block, divided by its expectation; the deviation in units of its standard error in brackets.

Diagonal. Figure 4 shows $\sigma_{\text{mock}}/\sigma_{\text{thecov}}$ for all 174 elements. The ratios scatter about unity with an rms of 0.025, against 0.022 expected from sampling alone (the small excess is the positive correlation between neighbouring bins, $\rho \approx 0.3$, which makes runs of points move together); the mean variance ratio over all elements is 0.997. No trend with k is visible in any block, in particular none at high k in the monopole-auto blocks, where the control run below fails.

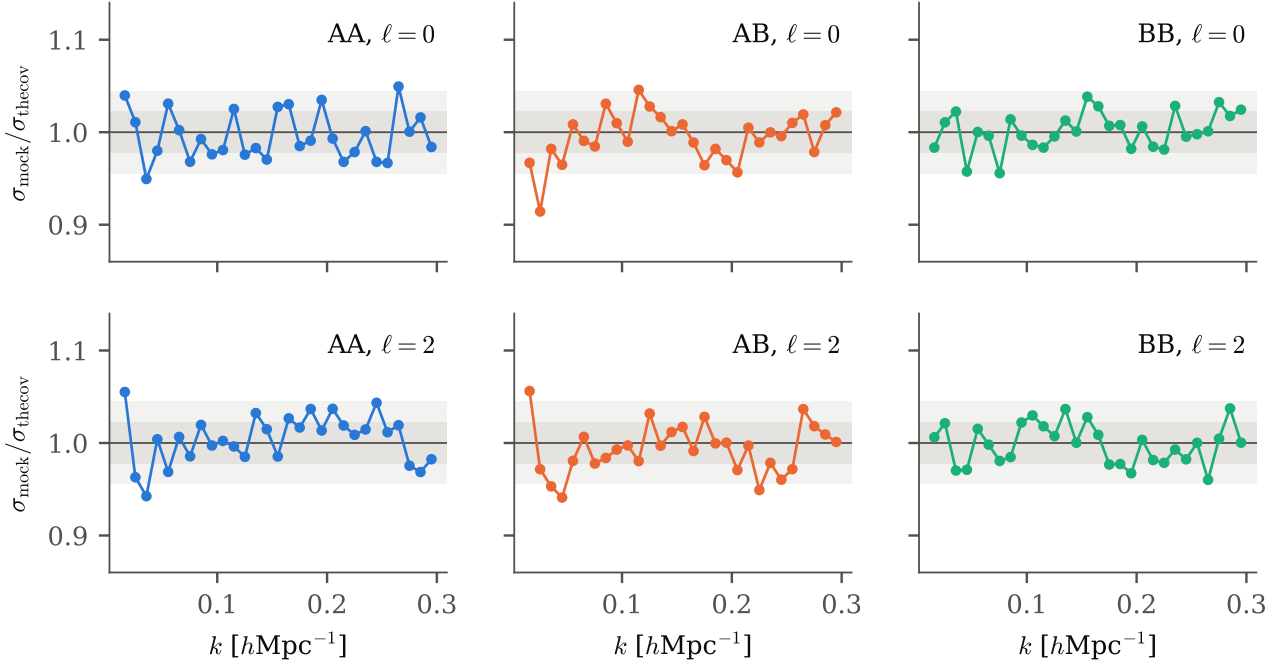


Figure 4: Ratio of the mock to the analytic standard deviation for every element of the data vector. Shaded: $\pm 1\sigma$ and $\pm 2\sigma$ sampling scatter for $N = 1000$.

Correlations. The full correlation matrices are compared in Figure 5. The analytic matrix reproduces the structure of the mocks: nearest-neighbour correlations $\rho \approx 0.3$ from the window, the coupling between $\ell = 0$ and $\ell = 2$ of the same spectrum, and the correlations between the auto- and cross-spectra, which carry the multi-tracer information. The normalised residuals z_{ij} have mean 0.035 and standard deviation 1.008; a fraction 0.27% of them exceed 3 in modulus, against 0.27% for a Gaussian.

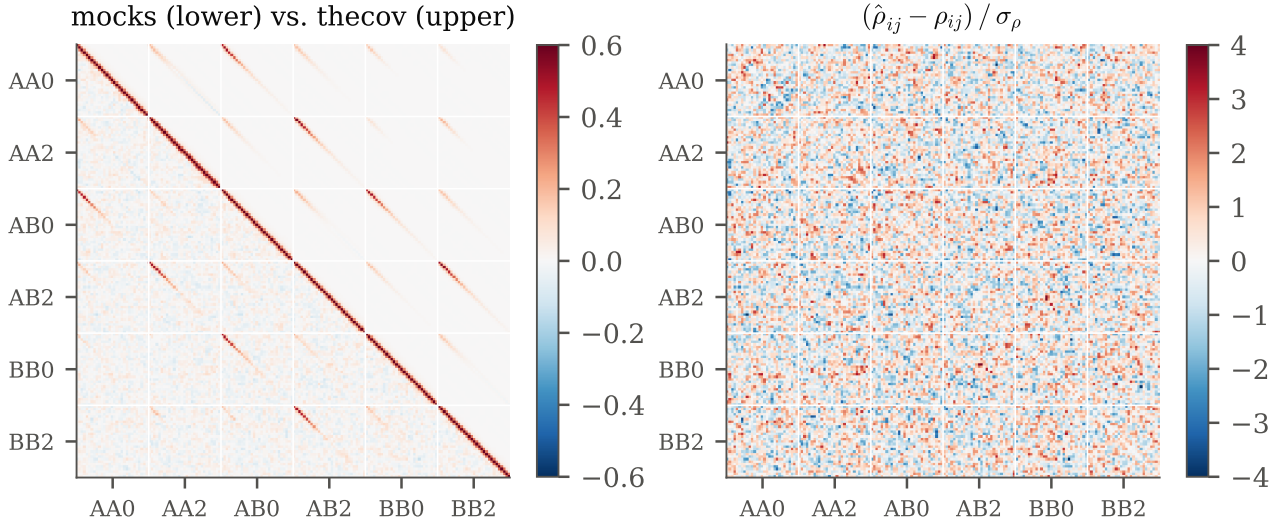


Figure 5: *Left*: correlation matrix of the mocks (lower triangle) and of `thecov` (upper triangle), ordered by spectrum, multipole and k bin. *Right*: the difference in units of its sampling error, z_{ij} .

5. Discussion

What is tested. With Gaussian mocks, the disconnected covariance is the complete answer, so a pass tests the window treatment and the two geometric approximations of Equation 1: the local plane-parallel line of sight — each pair sharing the direction of one of its points in a 30° cap, with redshift-space displacements along individual lines of sight — and the slowly varying window, tested on a footprint with holes, a completeness gradient and steep radial selections. Neither leaves a detectable imprint at the 0.3% precision of Equation 2, nor at the 2% per-element level of Figure 4. The same holds for the multi-tracer structure: the auto-cross correlations, which depend on the cross window W^{AB} and on the separate shot-noise windows of the two tracers, are reproduced as well as the auto-spectrum blocks.

The shot-noise convention matters. A control run on the same survey that subtracted the *expected* shot noise $(1 + \alpha)\alpha \sum_r w_r^2$ instead of the realised $\sum_g w_g^2 + \alpha^2 \sum_r w_r^2$ failed at $\langle \chi^2 \rangle = 178.03$ ($+7.1\sigma$). With the expected value, the self-pairs stay in $\hat{P}$ and the Poisson fluctuation of their sum adds to the monopole-auto covariance the fully correlated term

$$\Delta C_{ij} = \left[\int \bar{n} w^4 + 2(P_0(k_i) + P_0(k_j)) \int \bar{n}^2 w^4 \right] / I^2, \quad (3)$$

which is 2%, 10% and 19% of the AA diagonal at $k = 0.12, 0.22$ and 0.30 here, and absent from any Gaussian formula. Adding Equation 3 to `thecov` without any free parameter brings the control run to 175.67 and removes its high- k excess (Figure 6). With the realised shot noise — the convention of `pypower` and `jaxpower`, the estimators `thecov` is meant to be used with — the self-pairs are removed exactly and Equation 3 does not arise, which is what the present run confirms. We checked the premise directly: the variance over the mocks of $\sum_g w_g^2$ is 429 (A) and 344 (B), against a prediction of $330 + 116$ and $304 + 59$ from the Poisson term $\int \bar{n} w^4$ plus the clustering of the weighted counts, $\int \bar{n} w^2 \bar{n} w^2 \xi$ (ratios 0.96 and 0.95, ± 0.045).

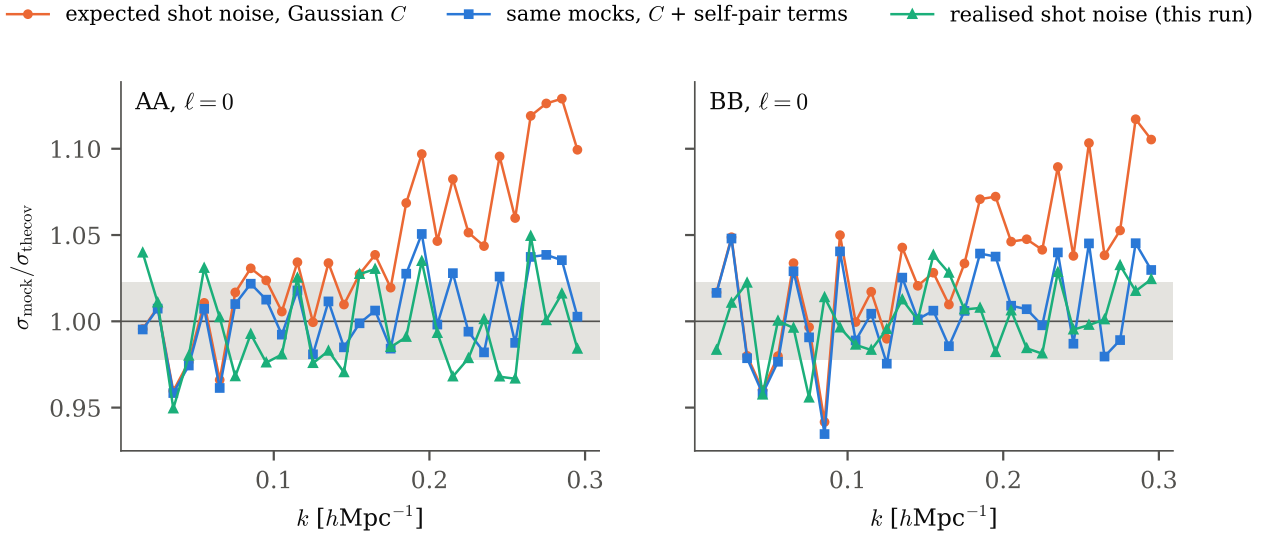


Figure 6: Monopole-auto blocks. *Orange*: control run with the expected shot noise, against the Gaussian `thecov` matrix. *Blue*: the same mocks against `thecov` plus the self-pair term Equation 3. *Aqua*: the present run (realised shot noise) against the Gaussian matrix. Shaded: $\pm 1\sigma$ sampling scatter.

Low k . The only feature suggestive of a departure is in the three lowest bins ($k = 0.015$ – 0.035), where the variance ratio averaged over the six blocks is 1.037, 0.966 and 0.941; over the three bins together it is 0.981 ± 0.015 (bootstrap over mocks), against 0.998 ± 0.004 for the remaining 26 bins. The deficit is 1.2σ from unity and not significant, but its sign and location are those expected from the integral constraint: the realised $\alpha = \sum w_d / \sum w_r$ forces the zero mode of the FKP field to vanish, which removes variance at $k \sim 1/R_{\text{survey}}$ and which Equation 1 does not model. An order-of-magnitude larger number of mocks, or the corresponding correction to Equation 1 (a Gaussian-level term involving the window integrated against $\bar{n}w$), would settle it.

Limits of the test. (i) The mocks are Gaussian: the trispectrum, super-sample covariance and the bispectrum and $P/\bar{n}^2$ discreteness terms are absent or negligible by construction, so the test says nothing about them. (ii) Redshift-space distortions are linear (Kaiser); there is no fingers-of-God damping. (iii) One box size (twice the survey extent) was used: the periodic images of the survey lie beyond $900 h^{-1}\text{Mpc}$, where the correlation function is negligible, and modes below $2\pi/L = 0.0035 h \text{Mpc}^{-1}$ are missing, both small effects that we did not test by varying the box. (iv) The normalisation was fixed; with the mesh-based normalisation of `pypower` and `jaxpower` the covariance must be rescaled by the ratio of normalisations (a few per cent, handled by `GaussianCovariance.set_normalization`), and the fluctuation of that normalisation adds a term below 10^{-3} of the diagonal for this survey. (v) The window functions carry Monte-Carlo noise from the random subsamples: two seeds differ by $\approx 10^{-2}$ per element of the covariance and by $\approx 2 \times 10^{-3}$ in $\langle \chi^2 \rangle / n$ for a subsample four times smaller than the one used here.

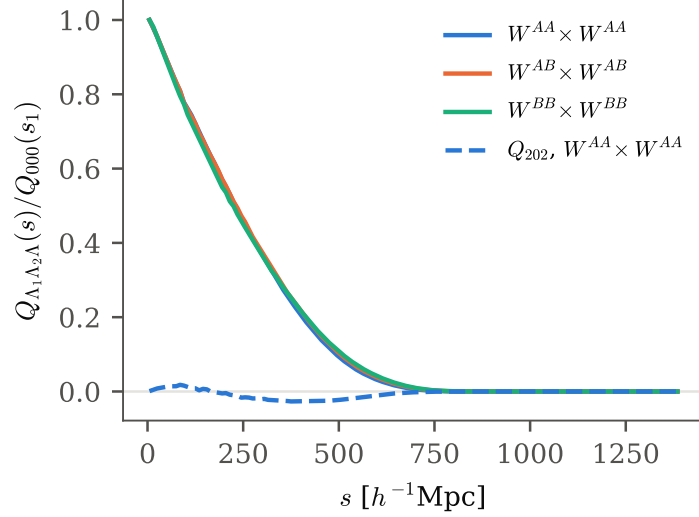


Figure 7: Monopole tripolar window functions $\mathcal{Q}_{000}(s)$ of the three clustering windows, and one anisotropic function, $\mathcal{Q}_{202}$, normalised to $\mathcal{Q}_{000}$ at the first separation bin.

6. Conclusion

For a two-tracer survey with a wide-angle, masked footprint, the separation-space Gaussian covariance reproduces the covariance of 1000 Gaussian mocks to the precision the mocks allow: $\langle \chi^2 \rangle$ within 0.7σ (0.25%), eigenvalues within the Marchenko–Pastur range of an exact matrix, every block within 1.5σ , and correlation residuals indistinguishable from sampling noise. Two conditions are essential and are enforced by the validation driver: the estimator must remove the self-pairs (realised shot noise, as in `pypower` and `jaxpower`), and the normalisation used by `thecov` must be the estimator’s. The remaining open question is the integral constraint at $k \lesssim 0.04 h \text{ Mpc}^{-1}$, where we see a 1.2σ variance deficit of the expected sign.

Reproducibility. Mocks and covariance: `python -m mocks.run_validation --n-mocks 1000 --grid 256 --box-factor 2 --kmax 0.30 --interlace --scheme tsc --amplitude 0.6 --out results_v5 (branch thecov2)`; figures and numbers: `reports/validation_v5/make_figures.py`.